

Portable Continuation via Cross-Machine Witnessed Image Exchange

A Deterministic, Artifact-First Protocol for Offline-Capable Systems

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Abstract

We present a portable, artifact-first protocol for cross-machine image generation and verification under deterministic replay constraints. The system enables one machine to declare a bounded rendering specification, a second machine to fulfill that specification under explicit constraints, and the originating machine to verify the returned artifact without reliance on network trust. The protocol is designed to operate across heterogeneous hardware environments and without dependency on centralized infrastructure. The result is a minimal continuation-control system for visual artifacts, supporting reproducibility, refusal-first execution, and portable verification.

1 Introduction

Contemporary systems assume that image generation, storage, and transmission occur within networked environments governed by centralized services. This assumption introduces ambiguity in provenance, execution authority, and reproducibility.

We investigate an alternative model:

A machine declares a rendering specification. A second machine fulfills the specification under deterministic constraints. The resulting artifact is returned with a verifiable witness bundle.

This replaces trust in infrastructure with verification of artifacts.

2 System Model

We define three roles:

- **Maverick (Operator Node)**: Declares rendering specifications and verifies results.
- **Babylon (Protocol Layer)**: Defines the structure of request and response packets.
- **Renderer (Worker Node)**: Executes rendering under constraints and emits artifacts.

2.1 Core Objects

Specification Packet

- Declares rendering constraints (geometry, elements, bounds)
- Defines required outputs
- Includes request identifier

Response Packet

- Contains artifact hash
- Declares execution outcome
- Binds output to request identifier

Run Bundle

- Artifact (image)
- SHA-256 hashes
- Environment snapshot
- Decision record
- Manifest and verification script

3 Finite-State Model

The system is modeled as a finite-state automaton:

$$Q = \{IDLE, DECLARED, TRANSMITTED, VALIDATED, EXECUTING, EMITTED, RETURNED, V\}$$

$$q_0 = IDLE$$

$$F = \{CLOSED\}$$

Transitions occur under explicit admissibility checks. Failure states are terminal and preserved.

4 Decision Model

The system enforces strict precedence:

$$HALT > INCOMPLETE > PASS$$

- **HALT**: Constraint violation or non-admissible state
- **INCOMPLETE**: Missing witnesses or unresolved conditions
- **PASS**: Artifact produced and verified

5 Deterministic Replay

Let S denote a specification and O the produced artifact.

$$O = R(S)$$

where R is a deterministic rendering operator.

Replay requires:

$$R(S) = O \Rightarrow \text{hash}(O') = \text{hash}(O)$$

Any deviation results in:

$$\text{decision} = \text{HALT or INCOMPLETE}$$

6 Portability Across Machines

The protocol is designed to function across machines with varying capabilities:

- differing CPU architectures
- differing OS environments
- differing library versions

Portability is achieved by:

- constraining specifications to admissible operations
- capturing environment state in the run bundle
- verifying outputs via hash equivalence rather than execution trace

7 Offline Operation

No network connectivity is required.

Transmission may occur via:

- local file transfer
- removable media
- direct device-to-device communication

The protocol treats the transmission channel as opaque. Only artifact integrity matters.

8 Experimental Demonstration

A deterministic navigation simulation was used as a rendering test case.

- Input: heading vector transitions
- Process: constrained propagation across discrete steps
- Output: trajectory image

The same specification produced identical geometry across machines, confirming replay stability.

Small perturbations in early parameters resulted in divergent large-scale paths, demonstrating sensitivity to initial conditions and the necessity of precise specification.

9 Discussion

This system replaces implicit trust with explicit verification.

Key properties:

- Artifact-first evaluation
- Deterministic replay
- Refusal-preserving execution
- Channel independence

The protocol is minimal yet extensible. While demonstrated for images, it generalizes to arbitrary artifact classes.

10 Conclusion

We have demonstrated a portable continuation-control protocol for cross-machine image exchange.

The system establishes that:

An action is not considered complete unless it can be reproduced and verified under declared constraints.

This approach enables a new class of offline-capable, verifiable systems independent of centralized infrastructure.

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